

ANSWER BOOK

I AGREE TO ABIDE BY THE SENATE POLICY ON CONDUCT OF ASSIGNMENTS, MID-TERM EXAM, AND FINAL EXAM

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Final Exam Question on FSet:

Q2a. Solution:

$$\begin{array}{lcl} X1 = [1, 2, 3] & | & A1 = 0.2 | 1 + 0.5 | 2 + 0.9 | 3 \\ X2 = [3, 4, 5, 6] & | & A2 = 0.2 | 3 + 0.5 | 4 + 0.9 | 5 + 0.2 | 6 \\ \\ y = f(x_1, x_2) & | & x1 + x2 \\ Y = [4, 5, 6, 7, 8, 9] & | & B = f(A1, A2) = ? \end{array}$$

The calculations for the two-dimensional to one-dimensional universal mapping is shown below:

k	x_1	$\mu_{X_1}(x_1)$	x_2	$\mu_{X_2}(x_2)$	y	$\min [\mu_{X_1}(x_1), \mu_{X_2}(x_2)]$	$\mu_B(y)$
1	1	0.2	3	0.2	4	0.2	0.2
2	1	0.2	4	0.5	5	0.2	0.2
3	1	0.2	5	0.9	6	0.2	0.5
4	1	0.2	6	0.2	7	0.2	0.5
5	2	0.5	3	0.2	5	0.2	0.2
6	2	0.5	4	0.5	6	0.5	0.5
7	2	0.5	5	0.9	7	0.5	0.5
8	2	0.5	6	0.2	8	0.2	0.9
9	3	0.9	3	0.2	6	0.2	0.5
10	3	0.9	4	0.5	7	0.5	0.5
11	3	0.9	5	0.9	8	0.9	0.9
12	3	0.9	6	0.2	9	0.2	0.2

Since there are two input membership functions, $\mu_{X_1}(x_1)$ and $\mu_{X_2}(x_2)$ their minimum membership values are taken as the input membership value.

From the above calculations, it is clear that there are ordered pairs given by:

- (a) $[x_1, x_2] = [1, 4]$ with membership function $[\mu_{X_1}(x_1), \mu_{X_2}(x_2)] = [0.2, 0.5]$ and $[x_1, x_2] = [2, 3]$ with membership function $[\mu_{X_1}(x_1), \mu_{X_2}(x_2)] = [0.5, 0.2]$ which are mapped to the same point $y = 5$.
- (b) $[x_1, x_2] = [1, 5]$ with membership function $[\mu_{X_1}(x_1), \mu_{X_2}(x_2)] = [0.2, 0.9]$, $[x_1, x_2] = [2, 4]$ with membership function $[\mu_{X_1}(x_1), \mu_{X_2}(x_2)] = [0.5, 0.5]$ and, $[x_1, x_2] = [3, 3]$ with membership function $[\mu_{X_1}(x_1), \mu_{X_2}(x_2)] = [0.9, 0.2]$ which are mapped to the same point $y = 6$.
- (c) $[x_1, x_2] = [1, 6]$ with membership function $[\mu_{X_1}(x_1), \mu_{X_2}(x_2)] = [0.2, 0.2]$, $[x_1, x_2] = [2, 5]$ with membership function $[\mu_{X_1}(x_1), \mu_{X_2}(x_2)] = [0.5, 0.9]$ and, $[x_1, x_2] = [3, 4]$ with membership function $[\mu_{X_1}(x_1), \mu_{X_2}(x_2)] = [0.9, 0.5]$ which are mapped to the same point $y = 7$.
- (d) $[x_1, x_2] = [2, 6]$ with membership function $[\mu_{X_1}(x_1), \mu_{X_2}(x_2)] = [0.5, 0.2]$ and $[x_1, x_2] = [3, 5]$ with membership function $[\mu_{X_1}(x_1), \mu_{X_2}(x_2)] = [0.9, 0.9]$ which are mapped to the same point $y = 8$.
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Thus, their respective maximum membership values must be taken. The fuzzy set B can be obtained from the extension principle as

$$B = 0.2 \mid 4 + 0.2 \mid 5 + 0.5 \mid 6 + 0.5 \mid 7 + 0.9 \mid 8 + 0.2 \mid 9$$

Thus, the sum of all the membership function values of the derived fuzzy set B is

$$\text{sum} = 0.2 + 0.2 + 0.5 + 0.5 + 0.9 + 0.2$$

$$\text{sum} = 2.5000$$

Q2b. Solution:

Based on your calculations in Q2a we can see that,

- The number of one-to-one mappings is two: for point $y = 4, 9$.
- The number of two-to-one mappings is two: for point $y = 5, 8$.
- The number of three-to-one mappings is two: for point $y = 6, 7$.

Therefore, all the statements are correct.

```

%{
% one-to-one mapping:
X1 = -3:3;
ux1 = [0.5 0.7 0.9 1 0.8 0.4 0.2];

k = 1:length(X1);

y = [];
for i = 1:length(X1)
    y(end+1) = X1(i)^2 - 5;
end

ub = [];
for i = 1:length(y)
    uxs = [];
    for j = 1:length(ux1)
        if y(i) == y(j)
            uxs(end+1) = ux1(j);
        end
    end
    ub(i) = max(uxs);
end

T = array2table([k' X1' ux1' y' ub'], ...
    "VariableNames", {'k' 'x1' 'ux1' 'y' 'ub'})

b = dictionary(y, ub);

output = 'B = ';

keys = b.keys;
values = b.values;

for i = 1:numel(keys)
    output = [output sprintf('%.1f | %d', values(i), keys(i))];

    if i < numel(keys)
        output = [output ' + '];
    end
end

disp(output);
sum = sum(b.values())
%}

%{ %}
% two-to-one mapping:
X1 = [1, 2, 3];
X2 = [3, 4, 5, 6];

A1 = [0.2, 0.5, 1.0];
A2 = [0.2, 0.5, 1.0, 0.2];

k = 1:length(X1)*length(X2);

x1 = [];
a1 = [];

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for i = 1:length(X1)
    for j = 1:length(X2)
        x1(end+1) = X1(i);
        a1(end+1) = A1(i);
    end
end

x2 = [];
a2 = [];
for i = 1:length(X1)
    for j = 1:length(X2)
        x2(end+1) = X2(j);
        a2(end+1) = A2(j);
    end
end

y = [];
for i = 1:length(X1)
    for j = 1:length(X2)
        y(end+1) = X1(i) + X2(j);
    end
end

minm = [];
for i = 1:length(A1)
    for j = 1:length(A2)
        minm(end+1) = min(A1(i), A2(j));
    end
end

ub = [];
for i = 1:length(y)
    mins = [];
    for j = 1:length(minm)
        if y(i) == y(j)
            mins(end+1) = minm(j);
        end
    end
    ub(i) = max(mins);
end

T = array2table([k' x1' a1' x2' a2' y' minm' ub'], ...
    "VariableNames", {'k' 'x1' 'ux1' 'x2' 'ux2' 'y' 'min(ux1, ux2)' 'ub'})

b = dictionary(y, ub);

output = 'B = ';

keys = b.keys;
values = b.values;

for i = 1:numel(keys)
    output = [output sprintf('%.1f | %d', values(i), keys(i))];

    if i < numel(keys)
        output = [output ' + '];
    end
end
end

```

```
disp(output);  
sum = sum(b.values())
```